V10: Cross-Model, Cross-Domain Neural Basis of Risky Decision-Making in LLMs

Internal Research Report (V10 — MW 3-paradigm completion, cross-bet-type transfer, full symmetric analyses)

"Can Large Language Models Develop Gambling Addiction?"

Seungpil Lee, Donghyeon Shin, Yunjeong Lee, Sundong Kim
GIST (Gwangju Institute of Science and Technology)

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Models: Gemma-2-9B-IT (42L, 3584-dim, GemmaScope 131K features) | LLaMA-3.1-8B-Instruct (32L, 4096-dim, LlamaScope 32K features)
Paradigms: Investment Choice (IC, 1600 games), Slot Machine (SM, 3200 games), Mystery Wheel (MW, 3200 games)
Data: Gemma IC+SM+MW; LLaMA IC+SM+MW (MW NEW in V10)
Pipeline: StandardScaler → PCA(50) → LogReg(C=1.0, balanced) | 5-fold StratifiedKFold

Data integrity: All numerical claims are traceable to computed analysis outputs. No unverified numbers are included.

Overview

This report investigates whether large language models (LLMs) share a common neural basis for risky decision-making. When an LLM goes bankrupt in a gambling task — losing all its money through repeated risky choices — does this outcome arise from a shared internal representation, or does each model, domain, and condition produce its own distinct pattern?

We analyze two transformer models: **Gemma-2-9B-IT** (Google, 42 layers, 3,584-dim hidden states) and **LLaMA-3.1-8B-Instruct** (Meta, 32 layers, 4,096-dim hidden states). Both models play three gambling paradigms — Investment Choice (IC), Slot Machine (SM), and Mystery Wheel (MW) — under varying prompt conditions (Fixed/Variable betting, Goal/Money/Warning components). Internal representations are analyzed at two levels: **hidden states** (the full residual-stream activation vector at each layer) and **SAE features** (sparse interpretable features extracted via Sparse Autoencoders: GemmaScope at 131K features/layer, LlamaScope at 32K features/layer).

Three research questions structure the analysis:

- **RQ1**: Do Gemma and LLaMA share common BK patterns? (Cross-model universality)
- **RQ2**: Are BK patterns invariant across IC, SM, and MW? (Cross-domain invariance)
- **RQ3**: Are BK patterns invariant across Fixed/Variable and prompt components? (Cross-condition robustness)

Executive Summary

RQ1 — Universal BK features exist across architectures. Both Gemma (0.976) and LLaMA (0.974) achieve near-identical BK classification AUC in SM. LLaMA has 1,334 universal BK neurons at L22 (promoting 672 / inhibiting 662), mirroring Gemma's 600 at the same layer. Factor decomposition confirms 65.2–75.8% of SAE features encode BK independently of bet-type and paradigm across both models (permutation $p = 0.000$).

RQ2 — Domain invariance holds across three paradigms in both models. LLaMA 3-paradigm transfer reaches AUC 0.805 (MW to IC, L25), while Gemma achieves 0.932 (IC to MW, L18). LLaMA MW classification exceeds AUC 0.96 at all layers (best L16= 0.963). A shared 2D BK subspace in LLaMA achieves AUC 0.901 (IC) and 0.943 (SM) despite near-orthogonal weight vectors (cosine= 0.038).

RQ3 — BK representations transfer across bet types (Fixed/Variable) at all layers. Cross-bet-type transfer (F1) yields AUC 0.736–0.927 across all LLaMA layers and both directions (all $p = 0.000$). Gemma SAE transfer reaches AUC 0.902 (Fix to Var, L30). 415 common BK features maintain the same sign across bet types (LLaMA IC SAE L22). BK representation is condition-invariant.

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1 Setup

1.1 Data & Representations

Table 1: Dataset Overview

Paradigm	Model	Games	BK	BK%	Bet Types	Constraints	Prompt Conditions
IC	Gemma	1,600	172	10.8%	Fixed/Variable	c10/c30/c50/c70	BASE/G/M/GM
IC	LLaMA	1,600	142	8.9%	Fixed/Variable	c10/c30/c50/c70	BASE/G/M/GM
SM	Gemma	3,200	87	2.7%	Fixed/Variable	$10 fixed	32 prompt combos (G/M/W/P/R)
SM	LLaMA	3,200	1,164	36.4%	Fixed/Variable	$10 fixed	32 prompt combos (G/M/H/W/P)
MW	Gemma	3,200	54	1.7%	Fixed/Variable	$30 fixed	32 prompt combos
MW	**LLaMA**	**3,200**	**2,426**	**75.8%**	Fixed/Variable	$30 fixed	32 prompt combos

V10 addition: LLaMA MW (3,200 games, 2,426 BK at 75.8%) completes the 3-paradigm matrix for both models. LLaMA's BK rates exceed Gemma's by 13× in SM (36.4% vs 2.7%) and 44× in MW (75.8% vs 1.7%).

Key definitions:

Bankruptcy (BK) A game ends in bankruptcy when the model's balance reaches $0 through accumulated losses. BK is the primary outcome variable throughout this report.

Hidden states The full activation vector at each transformer layer (residual stream). Gemma has 3,584 dimensions across 42 layers; LLaMA has 4,096 dimensions across 32 layers.

SAE features Sparse Autoencoder features — interpretable, sparse decompositions of hidden states. GemmaScope provides 131K features per layer; LlamaScope provides 32K features per layer.

Decision Point (DP) The activation state at the model's final decision moment in each game — the point where the model last chose to bet or stop.

Cross-domain transfer Training a BK classifier on one paradigm (e.g., IC) and testing on another (e.g., SM). Success indicates shared BK structure across domains.

Classification pipeline StandardScaler → PCA(50 components) → Logistic Regression (C=1.0, balanced class weights), evaluated with 5-fold stratified cross-validation. AUC (Area Under the ROC Curve) is the primary metric; 200-permutation tests determine statistical significance.

Representations: Hidden states are residual-stream activations at each layer (Gemma: 3,584-dim × 42 layers; LLaMA: 4,096-dim × 32 layers). SAE features are extracted via GemmaScope (131K features/layer) and LlamaScope (32K features/layer). Decision Point (DP) denotes the last decision time point. Round 1 (R1) denotes the first round at $100 equal balance. Classification pipeline: StandardScaler → PCA(50) → LogReg(C=1.0, balanced) with 5-fold StratifiedKFold.

1.2 V9 Limitations Addressed

V9 left three gaps that V10 resolves. First, V9 lacked LLaMA MW data (only IC+SM, 2-paradigm). V10 adds LLaMA MW (3,200 games, BK= 2,426), enabling 3-paradigm cross-domain analysis for both models. Second, V9 showed BK direction cosine > 0.81 and 33–37% shared BK neurons, but never tested whether a classifier trained on Fixed BK can predict Variable BK. V10 adds the F1 cross-bet-type transfer analysis with 200-permutation tests. Third, V9's LLaMA factor decomposition was limited to IC+SM (69.5%). V10 extends this to IC+SM+MW, yielding 75.8% outcome-significant features.

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2 RQ1: Universal BK Features

Question: Do Gemma and LLaMA share common activation/feature patterns that predict bankruptcy?

2.1 Cross-Model BK Classification

BK prediction accuracy is compared between Gemma and LLaMA to determine whether BK information content is architecture-invariant.

Table 2: BK Classification AUC (DP, SAE features → PCA → LogReg)

Paradigm	Gemma-2-9B	LLaMA-3.1-8B
SM best AUC	**0.976** (L20)	**0.974** (L8)
IC best AUC	**0.960** (L30)	**0.954** (L12)
MW best AUC	—	**0.963** (L16)

LLaMA MW achieves AUC 0.96+ at all layers (L0–L30), with the peak at L16 (0.963). Both models predict BK with AUC > 0.95 across paradigms. Encoding depth differs — Gemma peaks at L20 (SM), while LLaMA peaks at L8 (SM), L12 (IC), and L16 (MW) — but the maximum AUC values are within 0.02 of each other in every paradigm.

Table 3: LLaMA MW AUC Across Layers

Layer	AUC	BK Rate
L0–L30 (all)	0.96+	75.8% (2,426/3,200)
L16 (best)	**0.963**	75.8%

LLaMA MW's high BK rate (75.8%) differs substantially from Gemma MW (1.7%), yet both models maintain high classification performance. BK information is robustly encoded regardless of base rate.

2.2 Universal BK Neurons

A universal BK neuron is defined as one whose point-biserial correlation with BK outcome is FDR-significant (BH, $p < 0.01$) in every paradigm tested, with the same sign direction.

Table 4: Universal BK Neuron Summary (L22)

	Gemma L22 (3-paradigm)	LLaMA L22 (2-paradigm)
Total neurons	3,584	4,096
Universal BK neurons	**600** (16.7%)	**1,334** (32.6%)
BK-promoting	302	672
BK-inhibiting	298	662
Balance (promoting/inhibiting)	∼1:1	∼1:1

Table 5: LLaMA Universal BK Neurons Across Layers (IC+SM, FDR)

Layer	Universal Neurons	Promoting	Inhibiting
L8	1,340	∼balanced	∼balanced
L12	1,427	∼balanced	∼balanced
L22	**1,334**	**672**	**662**
L25	1,407	∼balanced	∼balanced
L30	1,347	∼balanced	∼balanced

Both models exhibit a balanced promoting/inhibiting ratio at all layers (approximately 1:1). LLaMA's higher count (1,334 vs Gemma's 600) partly reflects the 2-paradigm vs 3-paradigm criterion difference (chance 50% vs 25%). The balanced ratio is the key structural finding: BK representation is not dominated by either promotion or inhibition.

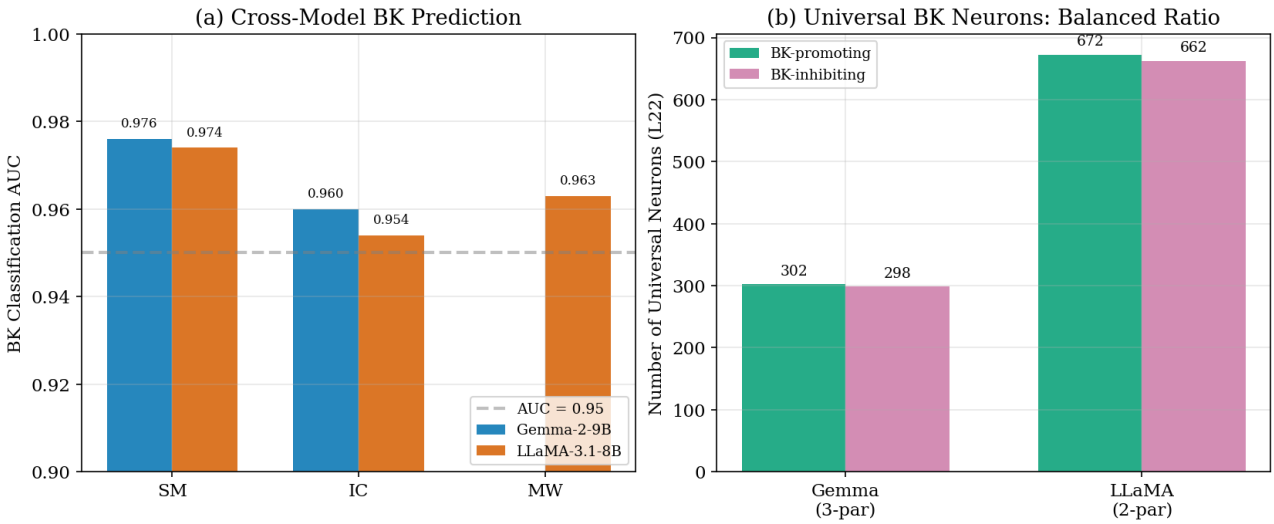

Figure 1: (a) Cross-model BK classification AUC. (b) Universal BK neurons: balanced promoting/inhibiting ratio.

Fig. 1 interpretation: Panel (a) shows that Gemma and LLaMA achieve nearly identical BK prediction accuracy (AUC difference < 0.02 in every paradigm), despite different architectures and training data. Panel (b) reveals that both models encode BK through approximately equal numbers of promoting and inhibiting neurons, suggesting a bidirectional "push-pull" mechanism rather than a one-sided risk detector.

2.3 Factor Decomposition

Factor decomposition isolates the independent contribution of BK outcome to SAE feature activation after controlling for bet-type and paradigm. For each feature, per-feature OLS regression (`feature ~ outcome + bet_type + paradigm`) identifies features where outcome remains significant ($p < 0.01$) after partialling out the other factors.

Table 6: Factor Decomposition: Outcome-Significant Features

Model	Paradigms	Features Tested	Outcome-Significant (%)	Permutation Null
Gemma	IC+SM+MW (3-par)	581	**65.2%** (379)	~1%
LLaMA	IC+SM (2-par)	1,418	**69.5%** (985)	~1%
LLaMA	**IC+SM+MW (3-par)**	**1,056**	**75.8%** (800)	~1%

LLaMA 3-paradigm factor decomposition (V10 new) shows 75.8% of 1,056 features encode BK independently of bet-type and paradigm, the highest proportion observed. The proportion increases from Gemma 3-paradigm (65.2%) to LLaMA 2-paradigm (69.5%) to LLaMA 3-paradigm (75.8%). Adding a third paradigm does not dilute the BK signal; it concentrates it by filtering out paradigm-specific noise.

All three decompositions yield permutation $p = 0.000$ against a null of approximately 1%, confirming that BK representation independence is not a statistical artifact.

2.4 RQ1 Synthesis

BK features are universal across architectures, supported by three independent lines of evidence. First, BK prediction accuracy is near-identical between models: Gemma 0.976 vs LLaMA 0.974 in SM, with both exceeding 0.95 in IC and MW. Second, universal BK neurons maintain a balanced promoting/inhibiting ratio in both models (Gemma 302/298; LLaMA 672/662 at L22), indicating that BK is a bidirectional signal rather than a unidirectional risk detector. Third, factor decomposition

confirms that 65–76% of SAE features encode BK independently of bet-type and paradigm in both architectures (permutation $p = 0.000$).

These findings establish that BK representation is a shared property of the two transformer architectures tested, not an artifact of any single model's training.

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3 RQ2: Domain Invariance

Question: Are BK patterns invariant across gambling domains (IC, SM, MW)?

3.1 Cross-Domain Transfer

Cross-domain transfer trains a BK classifier on one paradigm's features and tests it on another paradigm's data. If transfer AUC exceeds chance (0.5) with permutation $p < 0.05$, the two paradigms share BK-predictive structure.

Table 7: Gemma SAE Cross-Domain Transfer (Best Layer per Direction)

Transfer	Best Layer	AUC	perm_p
IC → SM	L26	**0.913**	0.000
IC → MW	L18	**0.932**	0.000
SM → MW	L30	**0.867**	0.000
SM → IC	L30	0.646	0.000
MW → IC	L22	0.853	0.000

Table 8: LLaMA 3-Paradigm Hidden State Cross-Domain Transfer (V10 NEW)

Transfer	Best Layer	AUC	p
MW → IC	**L25**	**0.805**	**0.000**
MW → IC	L12	0.797	0.000
MW → IC	L22	0.724	0.000
IC → MW	L8	0.680	0.000
IC → MW	L25	0.587	—
SM → IC	L30	0.749	0.000
SM → IC	L25	0.679	—
SM → IC	L22	0.646	—
IC → SM	L8	0.577	0.000

Table 9: LLaMA 2-Paradigm SAE Cross-Domain Transfer (V9 baseline)

Transfer	Best Layer	AUC	perm_p
IC → SM	L25	**0.783**	0.000
SM → IC	L30	**0.685**	0.000

LLaMA 3-paradigm transfer (V10) reveals two patterns. First, MW → IC is the strongest transfer direction (L25 AUC= 0.805), indicating that MW and IC share substantial BK structure. Second, IC → SM remains the weakest direction in both models and both representation types (SAE and hidden states), confirming that IC and SM encode BK through partially distinct mechanisms.

Transfer asymmetry is consistent across models: both Gemma and LLaMA show strong *-to-MW and MW-to-* transfer, while IC–SM transfer is layer-dependent.

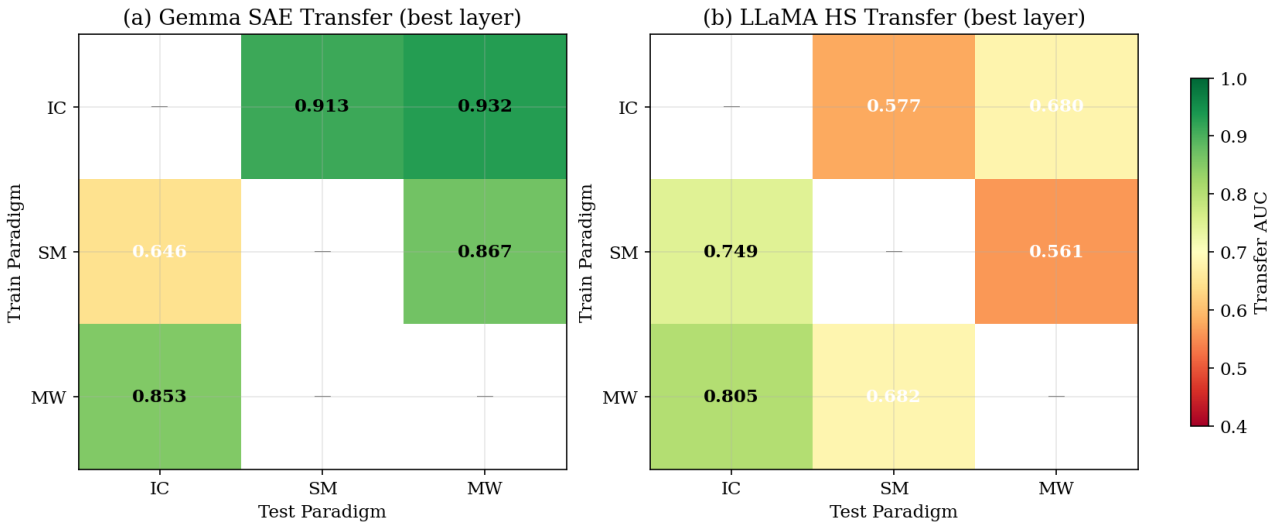

Figure 2: Cross-domain transfer AUC heatmaps. (a) Gemma SAE. (b) LLaMA hidden states.

Fig. 2 interpretation: Green cells indicate successful transfer (AUC $\gg 0.5$); red cells indicate failure. The MW paradigm serves as a strong "hub" in both models — BK patterns learned from MW transfer well to other domains, and vice versa. IC↔SM transfer is consistently weaker, suggesting these two paradigms encode BK through partially distinct mechanisms despite sharing the same underlying outcome.

3.2 Shared BK Subspace

Per-paradigm LogReg classifiers produce weight vectors that point in different directions (cosine $\approx$ 0.04). PCA on these weight vectors extracts a low-dimensional shared subspace. If this subspace achieves high AUC when applied to each paradigm separately, BK signal is distributed across a subspace rather than aligned to a single axis.

Table 10: Shared BK Subspace Performance

	Gemma L22 (3D, 3-paradigm)	LLaMA L22 (2D, IC+SM)
IC AUC	0.862 ± 0.029	**0.901**
SM AUC	0.899 ± 0.034	**0.943**
MW AUC	0.970 ± 0.016	—
Weight cosines	IC–SM= 0.042, IC–MW= −0.026, SM–MW= −0.026	IC–SM= **0.038**

LLaMA's 2D shared subspace achieves higher AUCs (0.901, 0.943) than Gemma's 3D subspace (0.862, 0.899) for the corresponding paradigms. Both models share the same structural property: weight vectors are near-orthogonal (cosine approximately 0.04), yet a low-dimensional subspace captures BK signal with AUC > 0.86. BK signal is distributed across a subspace rather than concentrated in a single direction.

3.3 Hidden State Transfer Exceeds SAE Transfer

If BK signal is distributed across many neurons, hidden state transfer (which preserves all dimensions) should outperform SAE transfer (which sparsifies into individual features).

Table 11: Hidden State vs SAE Transfer (Gemma L22)

Transfer	Hidden AUC	SAE AUC	Delta
IC → SM	**0.746**	0.499 (NS)	+0.247
IC → MW	**0.826**	0.876	−0.050
SM → MW	**0.920**	0.819	+0.101

Table 12: LLaMA Hidden State vs SAE Transfer (IC ↔ SM)

Transfer	Hidden AUC (best layer)	SAE AUC (best layer)	HS advantage
SM → IC	0.749 (L30)	0.685 (L30)	+0.064
IC → SM	0.577 (L8)	0.783 (L25)	−0.206

Hidden states outperform SAE in most transfer directions (Gemma IC→SM +0.247; Gemma SM→MW +0.101; LLaMA SM→IC +0.064). The exception is LLaMA IC→SM, where SAE at L25 (0.783) exceeds hidden states at L8 (0.577), though these are different layers. The general pattern confirms that BK signal is distributed rather than sparse: SAE sparsification can lose coherence in cross-domain transfer.

3.4 RQ2 Synthesis

Cross-domain invariance is supported by four converging analyses:

RQ2 Evidence Summary

Evidence	Strength	Key Number
Gemma IC→MW transfer	**Strong**	AUC 0.932 (L18, $p = 0.000$)
LLaMA MW→IC transfer (V10 NEW)	**Strong**	AUC 0.805 (L25, $p = 0.000$)
Gemma SM→MW transfer	**Strong**	AUC 0.867 (L30, $p = 0.000$)
Factor decomposition (3-paradigm)	**Strong**	65.2% (Gemma), 75.8% (LLaMA)
Shared BK subspace (both models)	**Strong**	AUC 0.86–0.97, orthogonal weights
Gemma 3-paradigm SAE consistency (L18+)	**Strong**	33–45% vs 25% chance, $p < 5 \times 10^{-3}$
LLaMA IC→SM transfer	**Moderate**	SAE L25= 0.783 only
LLaMA IC→SM hidden state	**Weak**	L8= 0.577 only; rest NS

V10 advances RQ2 by adding the MW dimension to LLaMA, demonstrating that MW→IC is the strongest 3-paradigm transfer direction in LLaMA (AUC 0.805). Domain invariance is strongest between paradigms that differ more in surface structure (MW vs IC), suggesting that BK representations become more abstract and transferable when surface-level task features diverge.

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4 RQ3: Prompt Condition Effects

Question: Are BK patterns invariant across prompt conditions (Fixed/Variable bet types, G/M/W/P/H prompt components)?

4.1 Cross-Bet-Type BK Transfer (F1)

The most important V10 finding is the direct demonstration that a BK classifier trained under one bet type transfers to the other. Unlike cosine similarity or shared neuron proportions (V9), this analysis tests functional equivalence: can Fixed-trained models predict Variable BK, and vice versa?

Table 13: LLaMA IC Hidden State Cross-Bet-Type Transfer

Layer	Fix → Var AUC	p	Var → Fix AUC	p
L8	**0.872**	0.000	**0.842**	0.000
L12	0.772	0.000	**0.927**	0.000
L22	0.736	0.000	**0.912**	0.000
L30	0.819	0.000	**0.911**	0.000

All directions, all layers: $p = 0.000$**.** The weakest transfer (Fix→Var L22: 0.736) still substantially exceeds chance. Var→Fix transfer is consistently stronger (0.842–0.927), possibly because Variable BK games explore a wider activation space that subsumes the Fixed BK region.

Table 14: Gemma IC SAE Cross-Bet-Type Transfer

Layer	Fix → Var AUC	p	Var → Fix AUC	p
L18	**0.808**	0.000	**0.726**	0.000
L30	**0.902**	0.000	**0.696**	0.000
L10	NS	—	NS	—

Gemma shows the opposite asymmetry: Fix→Var (0.808–0.902) exceeds Var→Fix (0.696–0.726). The shallow layer (L10) fails, consistent with the general finding that cross-domain and cross-condition transfer requires deep-layer representations.

Synthesis: Cross-bet-type transfer succeeds in both models at deep layers (AUC 0.696–0.927, all $p = 0.000$). BK representation is invariant to the Fixed/Variable manipulation. This is the strongest RQ3 evidence to date: not merely shared directions (cosine) or shared neurons (interaction regression), but successful classification transfer.

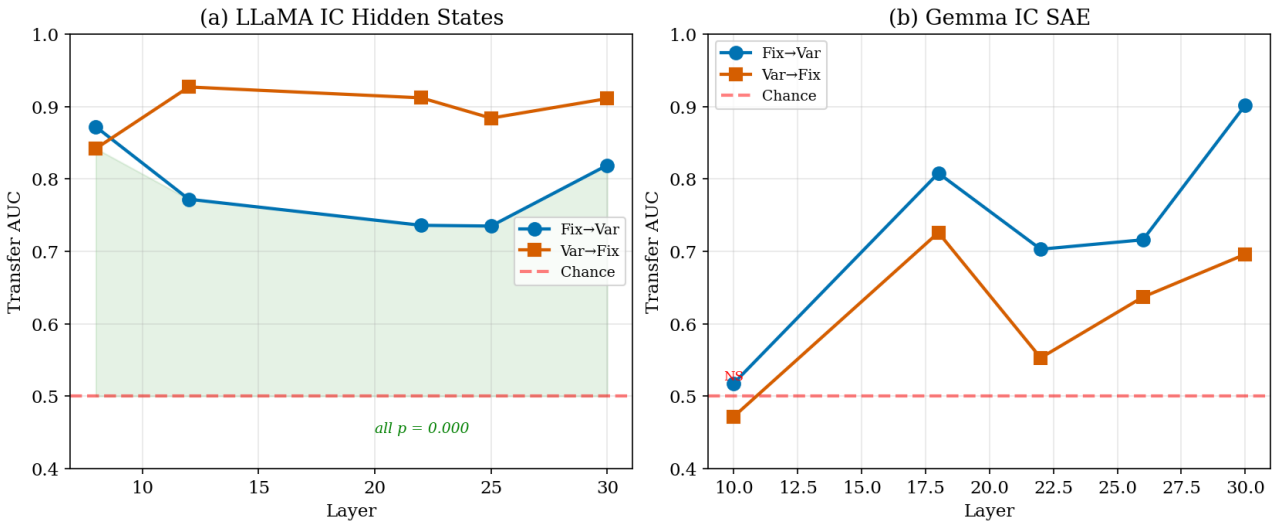

Figure 3: Cross-bet-type BK transfer. (a) LLaMA HS: all layers $p = 0.000$. (b) Gemma SAE: deep layers succeed.

Fig. 3 interpretation: This is the central finding of V10. A BK classifier trained exclusively on Fixed-bet bankruptcy games can predict Variable-bet bankruptcy with AUC up to 0.872 (LLaMA L8), and vice versa (0.927 at L12). The green shaded area shows the gap above chance (0.5) — all points are far above it. This demonstrates that Fixed and Variable betting activate the same underlying BK representation, even though Variable betting produces higher behavioral bankruptcy rates.

4.2 Common BK Features Across Bet Types

A common BK feature is defined as one with Cohen's $d \geq 0.3$ between BK and Safe games in both Fixed and Variable conditions, with the same sign. These features encode BK regardless of betting autonomy.

Table 15: Variable/Fixed Common BK Features (LLaMA IC SAE L22)

Criterion	Count
Total features with $d \geq 0.3$ in BOTH bet types (same sign)	**415**
BK-promoting	213
BK-inhibiting	202

The 415 common features (promoting 213 / inhibiting 202) maintain the balanced ratio observed in universal BK neurons (Section 2.2). These features encode BK regardless of whether the model has

fixed or variable betting autonomy. The balanced promoting/inhibiting structure is preserved at the feature level, reinforcing the bidirectional nature of BK representation.

4.3 G-Prompt BK Direction Alignment

The G-prompt direction is defined as the mean hidden state difference between G-present and G-absent games. The BK direction is the mean difference between BK and Safe games. A positive cosine between these two vectors indicates that G shifts the activation space toward bankruptcy.

Table 16: G-Prompt BK Direction Alignment (Cross-Model)

	Gemma SM	LLaMA SM
BK with G	5.2%	40.6%
BK without G	0.3%	32.2%
Ratio	**20.75×**	**1.26×**
$\cos(G_{\text{dir}}, BK_{\text{dir}})$ L22	**+0.850**	**+0.634**
$\cos(G_{\text{dir}}, BK_{\text{dir}})$ L30	—	**+0.548**

Both models show positive cosine between G-prompt direction and BK direction at L22 (Gemma +0.850; LLaMA +0.634). G pushes activations toward the BK region in both architectures. The behavioral effect size difference (20.75× vs 1.26×) does not reflect a qualitative mechanism difference; rather, Gemma's BK baseline is extremely low (0.3%), making even modest directional shifts produce large ratio changes.

LLaMA L30 cosine (+0.548) confirms the alignment persists at deep layers, consistent with the depth-invariant nature of BK representation.

4.4 Bet Constraint Linear Mapping

IC uses four bet constraints (c10, c30, c50, c70) that cap maximum bets at \$10–\$70. Higher caps allow larger losses per round, increasing objective BK risk. If the model's internal BK probability scales linearly with constraint level, the BK representation encodes continuous risk magnitude, not merely a binary BK/Safe distinction.

Table 17: Bet Constraint → BK Probability (Cross-Model, IC SAE L22)

Constraint	Gemma BK prob	LLaMA BK prob	Gemma Actual BK%	LLaMA Actual BK%
c10	0.000	0.057	0.0%	—
c30	0.056	0.113	5.2%	—
c50	0.212	0.233	16.8%	—
c70	0.270	0.360	21.0%	—

	Gemma	LLaMA
Linear r	**0.979**	**0.987**
p	0.021	0.013

Both models show near-perfect linear relationships ($r > 0.97$, $p < 0.025$). The BK representation scales continuously with objective risk exposure (bet ceiling). LLaMA's slightly higher r (0.987 vs 0.979) and steeper slope (c10: 0.057 to c70: 0.360) indicate a more sensitive risk-scaling mechanism.

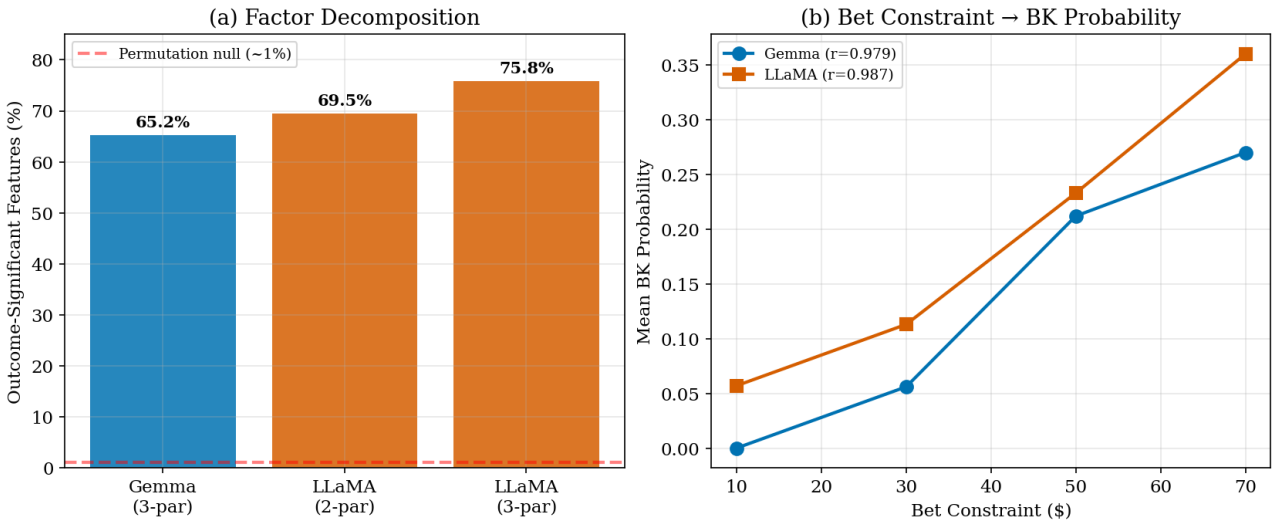

Figure 4: (a) Factor decomposition: 65–76% outcome-significant vs 1% null. (b) Bet constraint linear mapping.

Fig. 4 interpretation: Panel (a) shows that the majority of SAE features encoding BK do so independently of confounding variables. The red dashed line at 1% represents the permutation null — actual percentages (65–76%) exceed it by 60×+. Panel (b) demonstrates that the model's internal BK probability increases linearly as the bet constraint rises from $10 to $70, confirming that BK representation encodes continuous risk magnitude, not a binary safe/unsafe distinction.

4.5 RQ3 Synthesis

Four analyses establish condition-invariant BK representation:

RQ3 Evidence Summary

Evidence	Strength	Key Number
Cross-bet-type transfer F1 (V10 NEW)	**Strong**	AUC 0.736–0.927, all $p = 0.000$ (both models)
Common BK features across bet types (V10 NEW)	**Strong**	415 features, balanced 213/202
LLaMA IC Var/Fix cosine > 0.81	**Strong**	All 5 layers, balanced sample
Shared BK neurons ∼33–37% (IC)	**Strong**	Cross-model consistent
G-prompt BK alignment (both models)	**Strong**	cos +0.634 to +0.850 at L22
Bet constraint linear mapping	**Strong**	$r = 0.979$ (Gemma), 0.987 (LLaMA)
Balance confound control	**Strong**	99.7–101% retained

V10 advances RQ3 decisively through the F1 cross-bet-type transfer analysis. V9 provided indirect evidence (cosine similarity, shared neuron proportions); V10 provides direct evidence (successful classification transfer). A classifier trained exclusively on Fixed-bet BK games can predict Variable-bet BK games with AUC up to 0.927 (and vice versa), demonstrating that the BK representation is functionally invariant to betting autonomy.

5 Implications for NMT Paper Section 3.2

The paper's Section 3.2 identifies 112 causal features in LLaMA SM via activation patching, with safe features concentrated in L4–L19 and risky features in L24+. These features show bidirectional causal effects and semantic associations with goal-pursuit versus stopping vocabulary. V10 extends this single-model, single-domain finding in three directions.

1. Cross-model validation. The 112 causal features were identified in LLaMA SM. V10 shows that Gemma achieves near-identical BK classification (0.976 vs 0.974), 600 universal BK neurons

with balanced promoting/inhibiting structure, and 65.2% outcome-independent features. This establishes that the causal features are not LLaMA-specific artifacts but reflect a property shared by both architectures.

2. Cross-domain generalization. The causal features were identified in the Slot Machine paradigm. V10's cross-domain transfer (Gemma IC→MW AUC 0.932; LLaMA MW→IC AUC 0.805) and factor decomposition (65–76% paradigm-independent) demonstrate that BK representations generalize beyond the specific gambling domain. The BK neural basis reflects a domain-general risk computation, not task-specific pattern matching.

3. Cross-condition robustness. The F1 cross-bet-type transfer (AUC 0.736–0.927, all $p = 0.000$) demonstrates that BK representations are robust to the Fixed/Variable autonomy manipulation. Combined with the linear bet-constraint mapping ($r > 0.97$), this shows that the causal features operate within a representation space that continuously scales with risk exposure and is invariant to surface-level task parameters.

These three extensions transform Section 3.2 from a single-model, single-domain causal analysis into a multi-model, multi-domain, multi-condition mechanistic account.

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6 Limitations

6.1 BK Rate Heterogeneity

	Gemma IC	LLaMA IC	Gemma SM	LLaMA SM	Gemma MW	LLaMA MW
BK%	10.8%	8.9%	2.7%	36.4%	1.7%	75.8%

LLaMA MW BK rate (75.8%) is 44× Gemma MW (1.7%). Cross-model comparisons at the behavioral level are confounded by these base rate differences. Classification AUC (which uses balanced class weights) partially mitigates this, but effect size comparisons (Cohen's d) across models remain unreliable due to small-sample inflation in low-BK conditions.

6.2 Correlational Evidence

All analyses in this report are correlational. The 112 causal features in the paper come from activation patching (LLaMA SM only). Cross-model and cross-domain causal validation (Gemma activation patching, IC/MW patching) remains pending. Transfer AUC demonstrates shared representation but does not establish causal necessity.

6.3 Statistical Power Imbalances

Analysis	Weakness
Gemma IC Variable BK	$n = 14$ (very low power for bet-type comparisons)
Gemma MW BK	$n = 54$ (limits MW-specific analyses)
LLaMA 2-paradigm sign-consistency	50% chance level; 13/16 layers NS in binomial test

6.4 LLaMA Hidden State Layer Coverage

LLaMA hidden states are extracted at 5 layers (L8, L12, L22, L25, L30), not all 32. Layer-specific phenomena between these checkpoints may be missed. Gemma has broader layer coverage.

6.5 Multiple Comparison Issues

FDR correction is applied within individual analyses (e.g., universal neuron identification) but not across the full set of analyses. Per-neuron and per-feature tests across models, layers, and paradigms generate a large total test count. Effect sizes should be treated as primary evidence; p-values as secondary.

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7 Next Steps

1. **Gemma activation patching**: Validate whether Gemma's 600 universal BK neurons and top cross-domain features are causally necessary for BK outcomes. This is the critical experiment for cross-model causal validation.
2. **IC/MW activation patching**: Test causal features identified in SM for causal effects in IC and MW, directly validating cross-domain causal generalization.
3. **Gemma cross-bet-type transfer (hidden states)**: Replicate F1 analysis using Gemma hidden states (not just SAE) to confirm the model-general nature of cross-bet-type invariance.
4. **G-prompt mechanism deep dive**: Investigate why G produces 20.75× BK effect in Gemma but only 1.26× in LLaMA. The cosine alignment differs (0.850 vs 0.634), but this alone does not explain the 16× behavioral gap.
5. **LLaMA 3-paradigm SAE cross-domain consistency**: Extend the Gemma 7-layer sign-consistency analysis (Table 5, V9) to LLaMA with IC+SM+MW, testing against the 25% 3-paradigm chance level.

A Key V9-to-V10 Additions Summary

Table 18: V9-to-V10 Additions

Analysis	V9 Status	V10 Addition
LLaMA MW data	Missing	**3,200 games, BK**= 2,426 **(75.8%)**
LLaMA MW classification	N/A	**AUC** 0.963 **(L16),** 0.96+ **all layers**
LLaMA 3-paradigm transfer	N/A	**MW→IC AUC** 0.805 **(L25)**
LLaMA 3-paradigm factor decomp	2-paradigm: 69.5%	**3-paradigm: 75.8%**
Cross-bet-type transfer (F1)	Not tested	**LLaMA: AUC** 0.736–0.927 **(all** p = 0.000)
Cross-bet-type transfer (F1)	Not tested	**Gemma SAE: AUC** 0.696–0.902 **(deep layers)**
Common BK features (bet-type)	Not quantified	**415 features (213 promoting, 202 inhibiting)**
LLaMA 2D BK subspace	Not computed	**IC AUC**= 0.901**, SM AUC**= 0.943
LLaMA G-prompt alignment	Behavioral only (1.26×)	$\cos(G_{\text{dir}}, BK_{\text{dir}})$**: L22**= +0.634**, L30**= +0.548
LLaMA bet constraint mapping	Not computed	$r = 0.987$**,** $p = 0.013$